

CHAPTER

24

**ELECTRIC
POTENTIAL**

24-2 Electric Potential Energy

When an electrostatic force acts between two or more charged particles within a system of particles, we can assign an **electric potential energy** U to the system.

If the system changes its configuration from an initial state i to a different final state f , the electrostatic force does work W on the particles. From Eq. 8-1, we then know that the resulting change ΔU in the potential energy of the system is

$$\Delta U = U_f - U_i = -W. \quad (24-1)$$

As with other conservative forces, the work done by the electrostatic force is *path independent*. Suppose a charged particle within the system moves from point i to point f while an electrostatic force between it and the rest of the system acts on it. Provided the rest of the system does not change, the work W done by the force on the particle is the same for *all* paths between points i and f .

Sample Problem

Work and potential energy in an electric field

Electrons are continually being knocked out of air molecules in the atmosphere by cosmic-ray particles coming in from space. Once released, each electron experiences an electrostatic force $\vec{F}$ due to the electric field $\vec{E}$ that is produced in the atmosphere by charged particles already on Earth. Near Earth's surface the electric field has the magnitude $E = 150 \text{ N/C}$ and is directed downward. What is the change ΔU in the electric potential energy of a released electron when the electrostatic force causes it to move vertically upward through a distance $d = 520 \text{ m}$ (Fig. 24-1)?

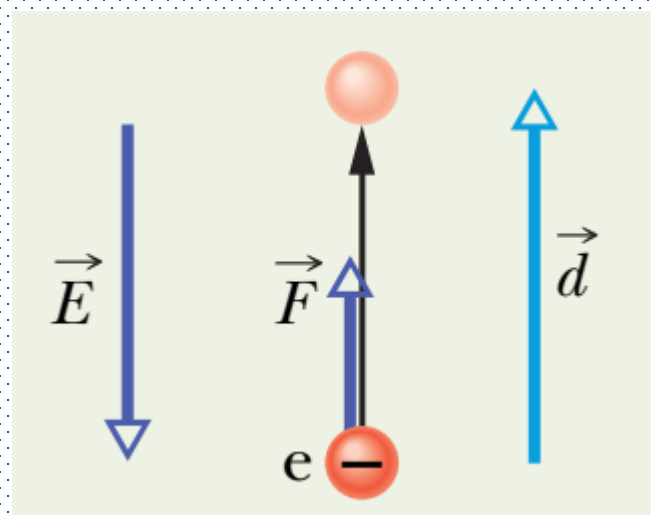
KEY IDEAS

(1) The change ΔU in the electric potential energy of the electron is related to the work W done on the electron by the electric field. Equation 24-1 ($\Delta U = -W$) gives the relation.

(2) The work done by a constant force $\vec{F}$ on a particle undergoing a displacement $\vec{d}$ is

$$W = \vec{F} \cdot \vec{d}. \quad (24-3)$$

(3) The electrostatic force and the electric field are related by the force equation $\vec{F} = q\vec{E}$, where here q is the charge of an electron ($= -1.6 \times 10^{-19}$ C).



Calculations:

$$W = q\vec{E} \cdot \vec{d} = qEd \cos \theta,$$

$$\begin{aligned} W &= (-1.6 \times 10^{-19} \text{ C})(150 \text{ N/C})(520 \text{ m}) \cos 180^\circ \\ &= 1.2 \times 10^{-14} \text{ J.} \end{aligned}$$

Equation 24-1 then yields

$$\Delta U = -W = -1.2 \times 10^{-14} \text{ J.}$$

This result tells us that during the 520 m ascent, the electric potential energy of the electron *decreases* by $1.2 \times 10^{-14} \text{ J}$.

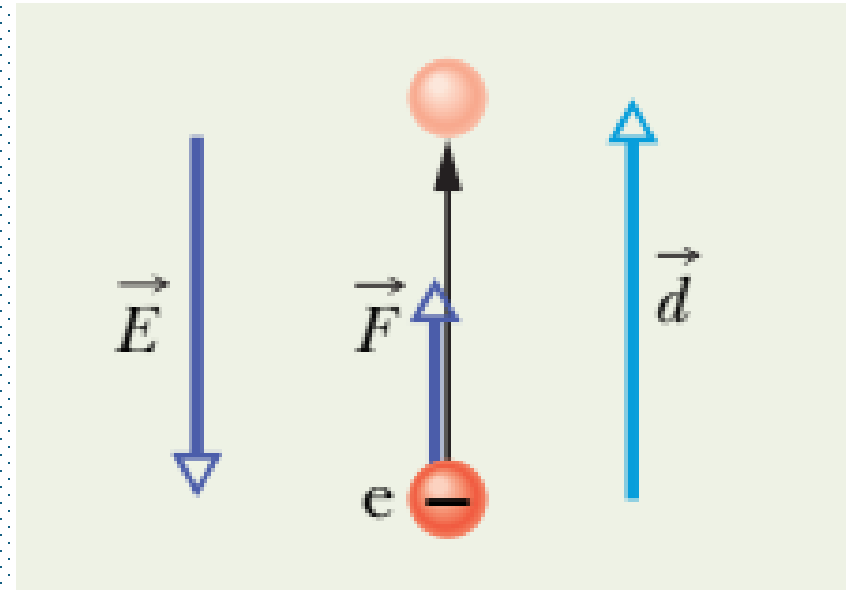
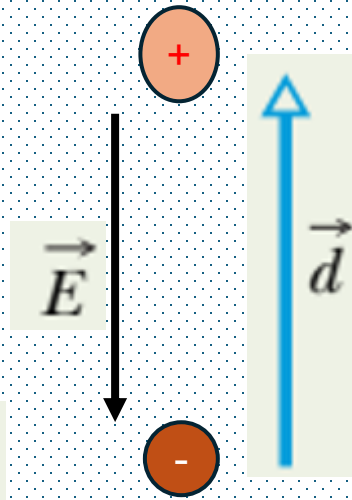
24-2 Electric Potential Energy

$$W = \vec{F} \cdot \vec{d}.$$

$$W = q\vec{E} \cdot \vec{d} = qEd \cos \theta,$$

$$\Delta U = -W$$

Poteitil Energy will decrease



24-3 Electric Potential

The potential energy of a charged particle in an electric field depends on the charge magnitude. However, the potential energy *per unit charge* has a unique value at any point in an electric field.

For an example of this, suppose we place a test particle of positive charge $1.60 \times 10^{-19} \text{ C}$ at a point in an electric field where the particle has an electric potential energy of $2.40 \times 10^{-17} \text{ J}$. Then the potential energy per unit charge is

$$\frac{2.40 \times 10^{-17} \text{ J}}{1.60 \times 10^{-19} \text{ C}} = 150 \text{ J/C.}$$

Next, suppose we replace that test particle with one having twice as much positive charge, 3.20×10^{-19} C. We would find that the second particle has an electric potential energy of 4.80×10^{-17} J, twice that of the first particle.

However, the potential energy per unit charge would be the same, still 150 J/C.

Thus, the potential energy per unit charge, which can be symbolized as U/q , is independent of the charge q of the particle we happen to use and is *characteristic only of the electric field* we are investigating. The potential energy per unit charge at a point in an electric field is called the **electric potential** V (or simply the **potential**) at that point. Thus,

$$V = \frac{U}{q}. \quad (24-5)$$

The *electric potential difference* ΔV between any two points i and f in an electric field is equal to the difference in potential energy per unit charge between the two points:

$$\Delta V = V_f - V_i = \frac{U_f}{q} - \frac{U_i}{q} = \frac{\Delta U}{q}. \quad (24-6)$$

Using Eq. 24-1 to substitute $-W$ for ΔU

$$\Delta V = V_f - V_i = -\frac{W}{q} \quad (\text{potential difference defined}). \quad (24-7)$$

The potential difference between two points is thus the negative of the work done by the electrostatic force to move a unit charge from one point to the other.

A potential difference can be positive, negative, or zero, depending on the signs and magnitudes of q and W .

If we set $U_i = 0$ at infinity as our reference potential energy, electric potential V must also be zero there.

$$V = -\frac{W_{\infty}}{q} \quad (\text{potential defined}), \quad (24-8)$$

$$\Delta V = V_f - V_i = \frac{U_f}{q} - \frac{U_i}{q} = \frac{\Delta U}{q}. \quad (24-6)$$

$$\Delta V = V_f - V_i = -\frac{W}{q} \quad (\text{potential difference defined}).$$

If we set $U_i = 0$ at infinity as our reference potential energy,

$$V = -\frac{W_\infty}{q}$$

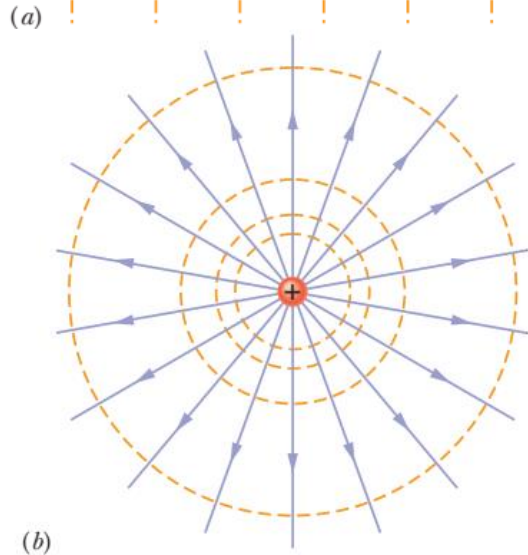
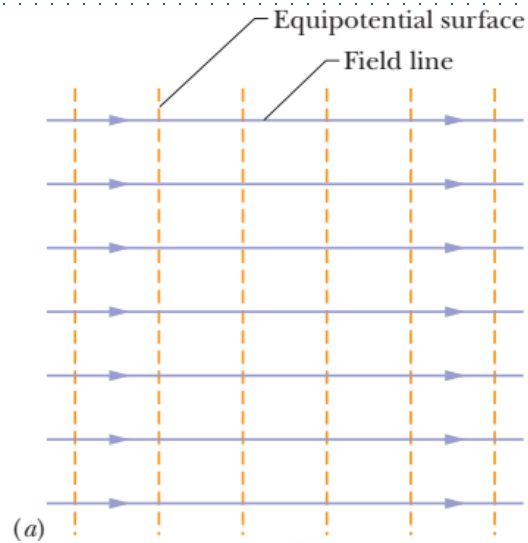
1 volt = 1 joule per coulomb.

$$\begin{aligned} 1 \text{ N/C} &= \left(1 \frac{\text{N}}{\text{C}}\right) \left(\frac{1 \text{ V} \cdot \text{C}}{1 \text{ J}}\right) \left(\frac{1 \text{ J}}{1 \text{ N} \cdot \text{m}}\right) \\ &= 1 \text{ V/m}. \end{aligned}$$

$$1 \text{ eV} = e(1 \text{ V})$$

$$= (1.60 \times 10^{-19} \text{ C})(1 \text{ J/C}) = 1.60 \times 10^{-19} \text{ J}.$$

24-4 Equipotential Surfaces



No work is done along this path on an equipotential surface.

Equal work is done along these paths between the same surfaces.

No work is done along this path that returns to the same surface.

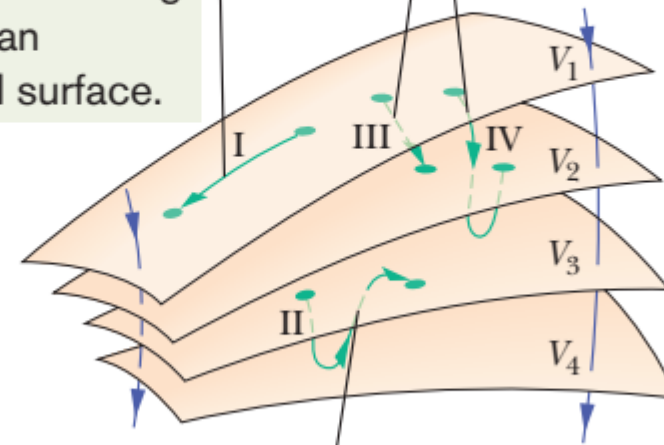


Fig. 24-2 Portions of four equipotential surfaces at electric potentials $V_1 = 100$ V, $V_2 = 80$ V, $V_3 = 60$ V, and $V_4 = 40$ V. Four paths along which a test charge may move are shown. Two electric field lines are also indicated.

24-5 Calculating the Potential from the Field

$$dW = \vec{F} \cdot d\vec{s}. \quad (24-15)$$

For the situation of Fig. 24-4, $\vec{F} = q_0 \vec{E}$ and Eq. 24-15 becomes

$$dW = q_0 \vec{E} \cdot d\vec{s}. \quad (24-16)$$

To find the total work W done on the particle by the field as the particle moves from point i to point f , we sum—via integration—the differential works done on the charge as it moves through all the displacements $d\vec{s}$ along the path:

$$W = q_0 \int_i^f \vec{E} \cdot d\vec{s}. \quad (24-17)$$

If we substitute the total work W from Eq. 24-17 into Eq. 24-7, we find

$$V_f - V_i = - \int_i^f \vec{E} \cdot d\vec{s}. \quad (24-18)$$

If we set potential $V_i = 0$,

$$V = - \int_i^f \vec{E} \cdot d\vec{s},$$

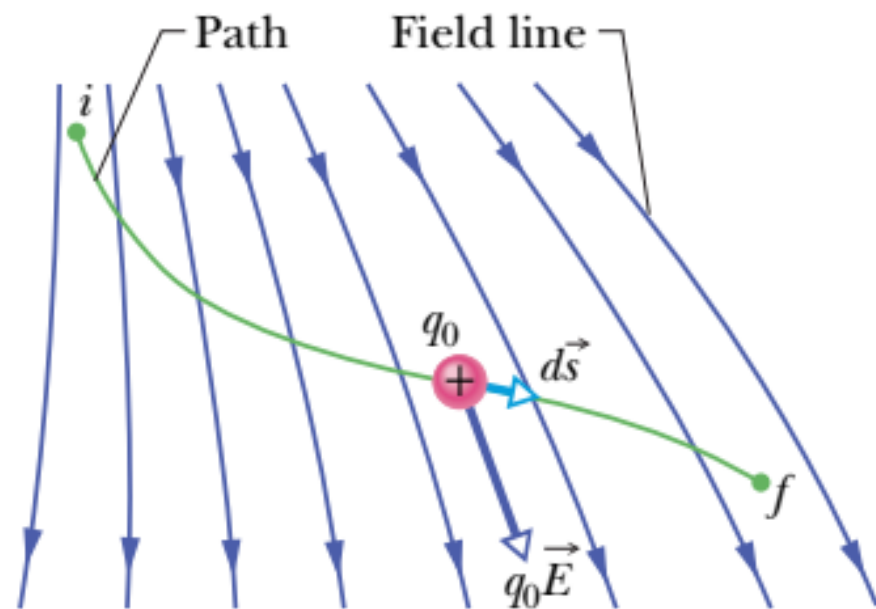


Fig. 24-4 A test charge q_0 moves from point i to point f along the path shown in a nonuniform electric field. During a displacement $d\vec{s}$, an electrostatic force $q_0\vec{E}$ acts on the test charge. This force points in the direction of the field line at the location of the test charge.

24-6 Potential Due to a Point Charge

$$\vec{E} \cdot d\vec{s} = E \cos \theta ds. \quad (24-22)$$

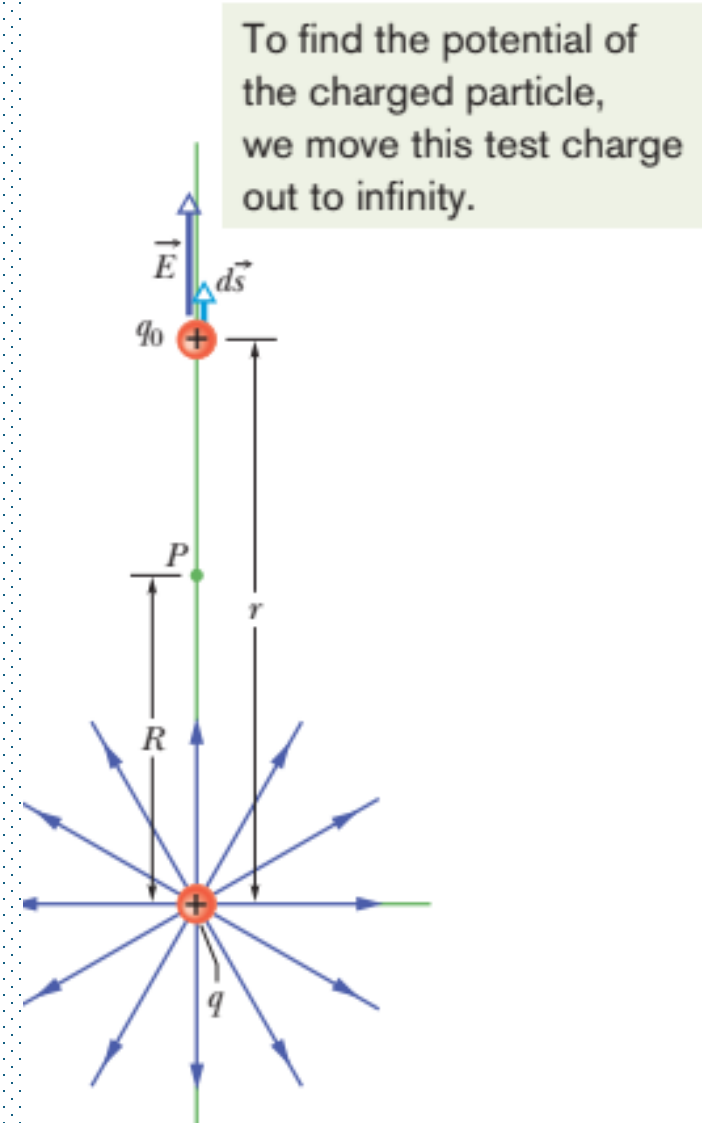
$$V_f - V_i = - \int_R^\infty E dr. \quad (24-23)$$

Next, we set $V_f = 0$ (at ∞) and $V_i = V$ (at R). Then, for the magnitude of the electric field at the site of the test charge, we substitute from Eq. 22-3:

$$E = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2}. \quad (24-24)$$

With these changes, Eq. 24-23 then gives us

$$0 - V = - \frac{q}{4\pi\epsilon_0} \int_R^\infty \frac{1}{r^2} dr = \frac{q}{4\pi\epsilon_0} \left[\frac{1}{r} \right]_R^\infty$$



$$= -\frac{1}{4\pi\epsilon_0} \frac{q}{R}. \quad (24-25)$$

Solving for V and switching R to r , we then have

$$V = \frac{1}{4\pi\epsilon_0} \frac{q}{r} \quad (24-26)$$

24-9 Potential Due to a Continuous Charge Distribution

When a charge distribution q is continuous (as on a uniformly charged thin rod or disk), we cannot use the summation of Eq. 24-27 to find the potential V at a point P . Instead, we must choose a differential element of charge dq , determine the potential dV at P due to dq , and then integrate over the entire charge distribution.

Let us again take the zero of potential to be at infinity. If we treat the element of charge dq as a point charge, then we can use Eq. 24-26 to express the potential dV at point P due to dq :

$$dV = \frac{1}{4\pi\epsilon_0} \frac{dq}{r} \quad (\text{positive or negative } dq). \quad (24-31)$$

Here r is the distance between P and dq . To find the total potential V at P , we integrate to sum the potentials due to all the charge elements:

$$V = \int dV = \frac{1}{4\pi\epsilon_0} \int \frac{dq}{r}. \quad (24-32)$$

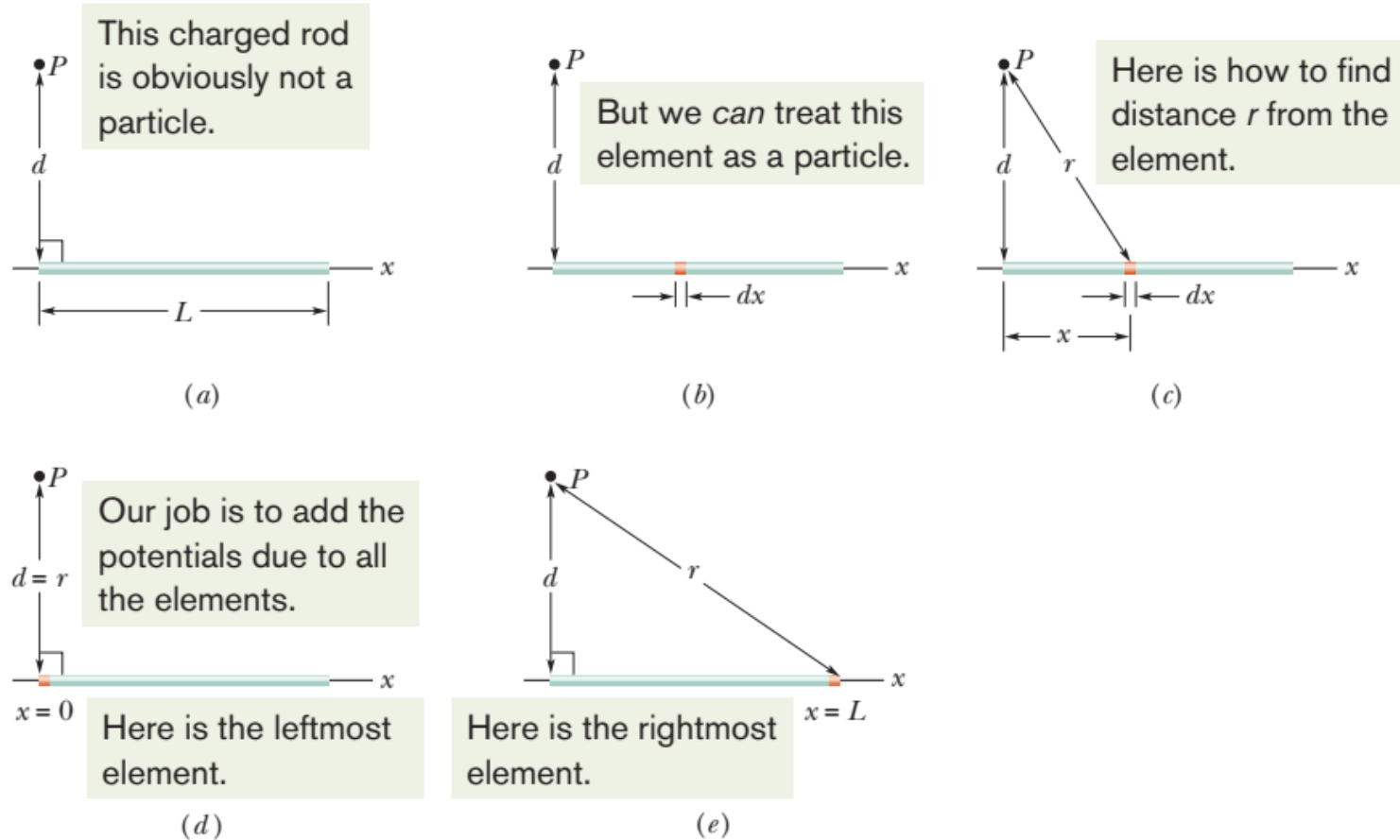
The integral must be taken over the entire charge distribution. Note that because the electric potential is a scalar, there are *no vector components* to consider in Eq. 24-32.

We now examine two continuous charge distributions, a line and a disk.

Line of Charge

In Fig. 24-12*a*, a thin nonconducting rod of length L has a positive charge of uniform linear density λ . Let us determine the electric potential V due to the rod

$$dq = \lambda dx.$$



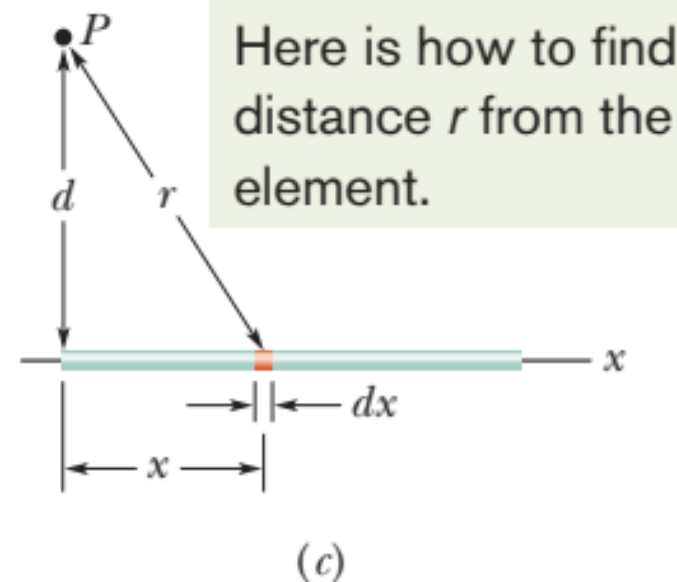
$$dV = \frac{1}{4\pi\epsilon_0} \frac{dq}{r} = \frac{1}{4\pi\epsilon_0} \frac{\lambda dx}{(x^2 + d^2)^{1/2}}.$$

$$V = \int dV = \int_0^L \frac{1}{4\pi\epsilon_0} \frac{\lambda}{(x^2 + d^2)^{1/2}} dx$$

$$= \frac{\lambda}{4\pi\epsilon_0} \int_0^L \frac{dx}{(x^2 + d^2)^{1/2}}$$

$$= \frac{\lambda}{4\pi\epsilon_0} \left[\ln\left(x + (x^2 + d^2)^{1/2}\right) \right]_0^L$$

$$= \frac{\lambda}{4\pi\epsilon_0} \left[\ln\left(L + (L^2 + d^2)^{1/2}\right) - \ln d \right].$$



$$\mathbf{17.} \quad \int \frac{dx}{\sqrt{x^2 + a^2}} = \ln(x + \sqrt{x^2 + a^2})$$

We can simplify this result by using the general relation $\ln A - \ln B = \ln(A/B)$.
We then find

$$V = \frac{\lambda}{4\pi\epsilon_0} \ln \left[\frac{L + (L^2 + d^2)^{1/2}}{d} \right]. \quad (24-35)$$

Charged Disk

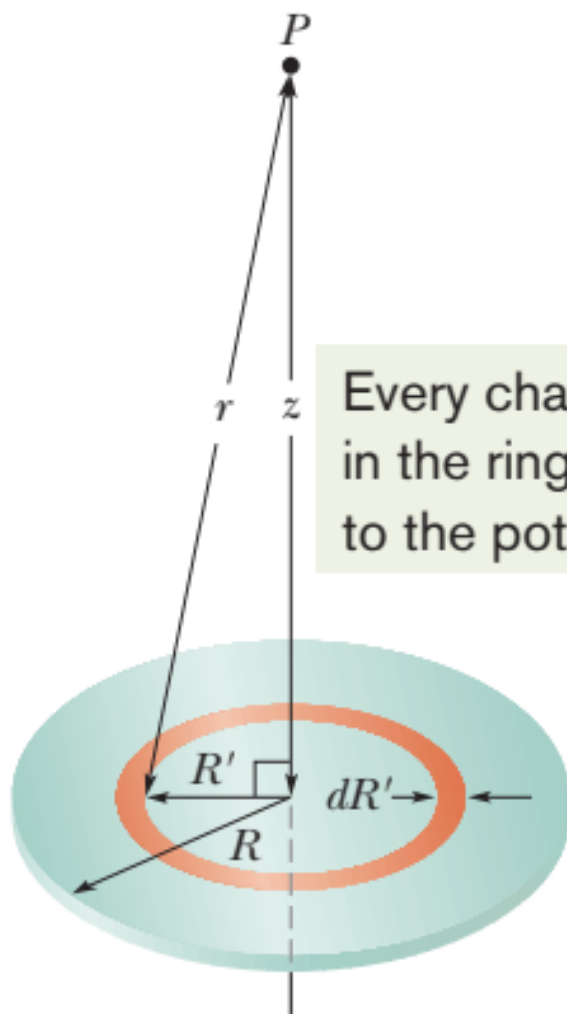
$$dq = \sigma(2\pi R')(dR'),$$

$$dV = \frac{1}{4\pi\epsilon_0} \frac{dq}{r} = \frac{1}{4\pi\epsilon_0} \frac{\sigma(2\pi R')(dR')}{\sqrt{z^2 + R'^2}}$$

We find the net potential at P by adding (via integration) the contributions of all the rings from $R' = 0$ to $R' = R$:

$$V = \int dV = \frac{\sigma}{2\epsilon_0} \int_0^R \frac{R' dR'}{\sqrt{z^2 + R'^2}}$$

$$= \frac{\sigma}{2\epsilon_0} (\sqrt{z^2 + R^2} - z).$$



Every charge element in the ring contributes to the potential at P .

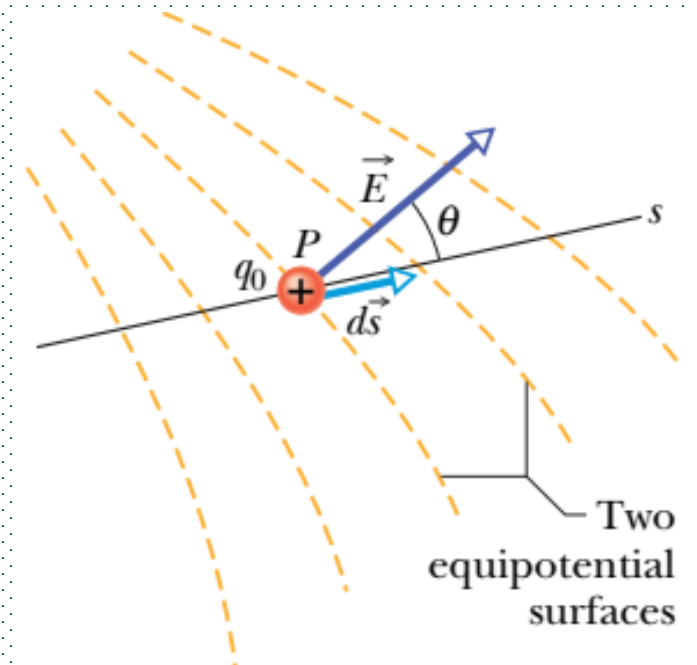
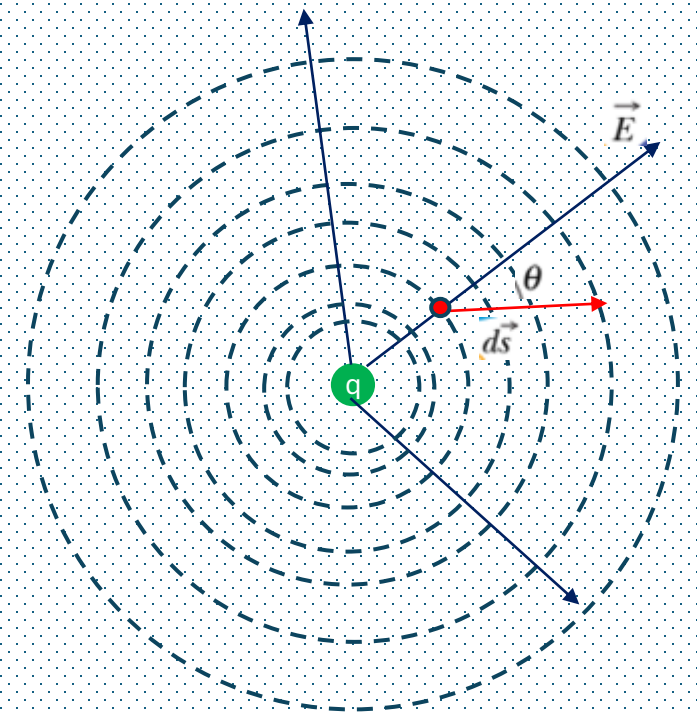
24-10 Calculating the Field from the Potential

$$-q_0 dV = q_0 E(\cos \theta) ds,$$

$$E \cos \theta = -\frac{dV}{ds}.$$

Since $E \cos \theta$ is the component of $\vec{E}$ in the direction of $d\vec{s}$,

$$E_s = -\frac{\partial V}{\partial s}.$$



If we take the s axis to be, in turn, the x , y , and z axes, we find that the x , y , and z components of $\vec{E}$ at any point are

$$E_x = -\frac{\partial V}{\partial x}; \quad E_y = -\frac{\partial V}{\partial y}; \quad E_z = -\frac{\partial V}{\partial z}. \quad (24-41)$$

$$E = -\frac{\Delta V}{\Delta s},$$

Sample Problem

Finding the field from the potential

The electric potential at any point on the central axis of a uniformly charged disk is given by Eq. 24-37,

$$V = \frac{\sigma}{2\epsilon_0} (\sqrt{z^2 + R^2} - z).$$

Starting with this expression, derive an expression for the electric field at any point on the axis of the disk.

KEY IDEAS

We want the electric field $\vec{E}$ as a function of distance z along the axis of the disk. For any value of z , the direction of $\vec{E}$ must be along that axis because the disk has circular symme-

try about that axis. Thus, we want the component E_z of $\vec{E}$ in the direction of z . This component is the negative of the rate at which the electric potential changes with distance z .

Calculation: Thus, from the last of Eqs. 24-41, we can write

$$\begin{aligned} E_z &= -\frac{\partial V}{\partial z} = -\frac{\sigma}{2\epsilon_0} \frac{d}{dz} (\sqrt{z^2 + R^2} - z) \\ &= \frac{\sigma}{2\epsilon_0} \left(1 - \frac{z}{\sqrt{z^2 + R^2}} \right). \end{aligned} \quad (\text{Answer})$$

This is the same expression that we derived in Section 22-7 by integration, using Coulomb's law.



Additional examples, video, and practice available at *WileyPLUS*

24-11 Electric Potential Energy of a System of Point Charges

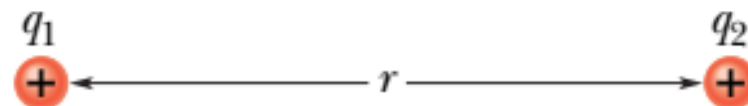


Fig. 24-15 Two charges held a fixed distance r apart.



The electric potential energy of a system of fixed point charges is equal to the work that must be done by an external agent to assemble the system, bringing each charge in from an infinite distance.

$$V = \frac{1}{4\pi\epsilon_0} \frac{q_1}{r}.$$

$$U = W = q_2 V = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r}. \quad (24-43)$$

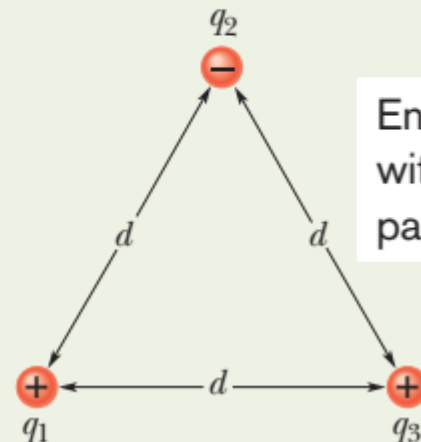
Sample Problem

Potential energy of a system of three charged particles

Figure 24-16 shows three point charges held in fixed positions by forces that are not shown. What is the electric potential energy U of this system of charges? Assume that $d = 12$ cm and that

$$q_1 = +q, \quad q_2 = -4q, \quad \text{and} \quad q_3 = +2q,$$

in which $q = 150$ nC.



Energy is associated with each pair of particles.

$$U_{12} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{d}.$$

$$W_{13} + W_{23} = U_{13} + U_{23} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_3}{d} + \frac{1}{4\pi\epsilon_0} \frac{q_2 q_3}{d}.$$

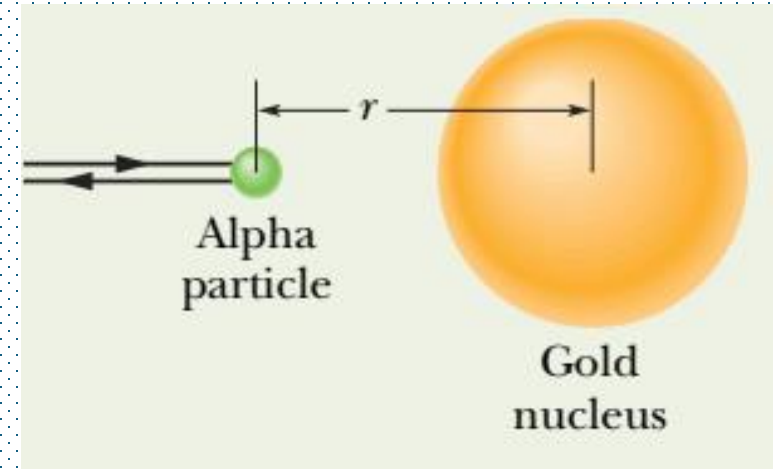
$$\begin{aligned} U &= U_{12} + U_{13} + U_{23} \\ &= \frac{1}{4\pi\epsilon_0} \left(\frac{(+q)(-4q)}{d} + \frac{(+q)(+2q)}{d} + \frac{(-4q)(+2q)}{d} \right) \\ &= -\frac{10q^2}{4\pi\epsilon_0 d} \\ &= -\frac{(8.99 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2)(10)(150 \times 10^{-9} \text{ C})^2}{0.12 \text{ m}} \\ &= -1.7 \times 10^{-2} \text{ J} = -17 \text{ mJ}. \end{aligned}$$

(Answer)

Sample Problem

Conservation of mechanical energy with electric potential energy

An alpha particle (two protons, two neutrons) moves into a stationary gold atom (79 protons, 118 neutrons), passing through the electron region that surrounds the gold nucleus like a shell and headed directly toward the nucleus (Fig. 24-17). The alpha particle slows until it momentarily stops when its center is at radial distance $r = 9.23$ fm from the nuclear center. Then it moves back along its incoming path.



Calculations: The principle of conservation of mechanical energy tells us that

$$K_i + U_i = K_f + U_f. \quad (24-44)$$

We know two values: $U_i = 0$ and $K_f = 0$. We also know that the potential energy U_f at the stopping point is given by the right side of Eq. 24-43, with $q_1 = 2e$, $q_2 = 79e$ (in which e is the elementary charge, $1.60 \times 10^{-19} \text{ C}$), and $r = 9.23 \text{ fm}$. Thus, we can rewrite Eq. 24-44 as

$$\begin{aligned} &= \frac{(8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2)(158)(1.60 \times 10^{-19} \text{ C})^2}{9.23 \times 10^{-15} \text{ m}} \\ &= 3.94 \times 10^{-12} \text{ J} = 24.6 \text{ MeV}. \quad (\text{Answer}) \end{aligned}$$

